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Effectiveness of digital infrared thermal imaging in detecting lower extremity deep venous thrombosis

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Purpose: The authors aimed to determine the effectiveness of infrared thermal imaging (IRTI) as a novel, noninvasive technique in adjunctive diagnostic screening for lower limb deep venous thrombosis (DVT).
Methods: The authors used an infrared thermal imaging sensor to examine the lower limbs of 64 DVT patients and 64 healthy volunteers. The DVT patients had been definitively diagnosed with either Doppler vascular compression ultrasonography or angiography. The mean area temperature (T_area) and mean linear temperature (T_line) in the region of interest were determined with infrared thermal imaging. Images were evaluated with qualitative pseudocolor analysis to verify specific color-temperature responses and with quantitative temperature analysis. Differences in T_area and T_line between the DVT limb and the nonaffected limb in each DVT patient and temperature differences (TDs) in T_area (TD_{area}) and T_line (TD_{line}) between DVT patients and non-DVT volunteers were compared.
Results: Qualitative pseudocolor analysis revealed visible asymmetry between the DVT side and non-DVT side in the presentation and distribution characteristics (PDCs) of infrared thermal images. The DVT limbs had areas of abnormally high temperature, indicating the presence of DVT. Of the 64 confirmed DVT patients, 62 (96.88%) were positive by IRTI detection. Among these 62 IRTI-positive cases, 53 (82.81%) showed PDCs that agreed with the DVT regions detected by Doppler vascular compression ultrasonography or angiography. In nine patients (14.06%), IRTI PDCs did not definitively agree with the DVT regions established with other testing methods, but still correctly indicated the DVT-affected limb. There was a highly significant difference between DVT and non-DVT sides in DVT patients ($P < 0.01$). The TD_{area} and TD_{line} in non-DVT volunteers ranged from 0.19 ± 0.15 °C to 0.21 °C ± 0.17 °C; those in DVT patients ranged from 0.86 °C ± 0.71 °C to 1.03 °C ± 0.79 °C ($P < 0.01$).
Conclusions: Infrared thermal imaging can be effectively used in DVT detection and adjunctive diagnostic screening because of its specific infrared PDCs and TDs values. © *2015 American Association of Physicists in Medicine.* [http://dx.doi.org/10.1118/1.4907969]

Key words: infrared thermal imaging (IRTI), deep venous thrombosis (DVT), presentation and distribution characteristics (PDCs), adjunctive diagnostic screening

1. INTRODUCTION

Venous thromboembolism includes pulmonary thromboembolism (PTE) and deep venous thrombosis (DVT). DVT alone is not life-threatening, but undiagnosed DVT may lead to potentially fatal complications, including PTE, which carries significant morbidity and mortality.[1–3] Approximately 300 000 people die from acute PTE in the United States each year; regrettably, many cases are diagnosed only at autopsy.[4] Ninety percent of PTE cases originate from lower limb DVT.[5] The incidence of lower extremity DVT is approximately 50 per 100 000 people per year.[1] Surprisingly, most cases are asymptomatic. Therefore, screening older individuals for DVT is worthwhile to avoid sudden onset of PTE.

Diagnosis of symptomatic DVT using Doppler vascular compression ultrasonography (CPUS) is well established.[6] The sensitivity and specificity of CPUS in diagnosing proximal DVT are 100% and 98%, respectively. Distal venous thrombosis is diagnosed with 94% sensitivity and 75% specificity

with CPUS.[7] Moreover, CPUS does not involve radiation exposure or nephrotoxic contrast agents.[8]

Body surface temperature is closely related to the volume of blood flow to the skin, and limb temperature depends on peripheral blood flow.[9] As a result, body temperature is useful in diagnosing various diseases;[9] thermography is one diagnostic method for assessing body temperature.[10] Thermography, the imaging of temperature distribution in the body, can also be used in diagnosing certain diseases, particularly in their early stages.[11] Several previous studies have confirmed the effectiveness of contact thermography in detecting various diseases, including DVT.[12–17] Infrared thermal imaging (IRTI) is a type of noncontact thermography that uses an uncooled infrared camera to capture the infrared radiation emitted by the body surface, thus measuring and recording body surface temperatures.[11] CPUS requires coupling medium contact, whereas IRTI uses remote capture without contact.[11] Furthermore, IRTI is simple and fast, whereas CPUS is more time-consuming. Additionally, IRTI is inexpensive and can be used to screen suspected DVT patients to avoid the costs of expensive and unnecessary medical examinations.

Studies have found a close relationship between skin surface temperature and limb blood flow.[10,18] Herry[19] found that surface temperatures in healthy individuals are typically symmetrical between the left and right sides of the body and that an area with markedly abnormal surface temperature revealed abnormal blood flow. Studies[10,20,21] have found that the temperature of the lower limbs primarily depends on peripheral vascular blood flow and that IRTI has diagnostic value in vascular diseases. However, these studies have mainly addressed disorders of peripheral arteries. Previous qualitative and quantitative analyses of IRTI have produced valuable information related to its use as an adjunctive diagnostic screening test for DVT.[22] Other IRTI researchers have pointed out that thermal imaging coupled with clinical findings can be used in adjunctive diagnostic screening for vascular disorders.[10,21,23]

We hypothesized that IRTI could diagnose and screen lower limb DVT of humans. Thus, the present study was intended to determine the clinical effectiveness of IRTI in adjunctive diagnostic screening of lower limb DVT in humans.

2. MATERIALS AND METHODS

The present clinical trial was approved by the Institutional Review Board of the First Affiliated Hospital of Guangzhou Medical University.

2.A. Subjects

DVT patients included in this study had been definitively diagnosed as having DVT with CPUS and/or angiography. Included non-DVT volunteers were confirmed to be without DVT with CPUS. The healthy volunteer group matched the DVT group in number, age, and gender.

2.B. Infrared thermal imaging acquisition and data analysis

The IRTI system is capable of precisely measuring and storing real-time thermal images with a resolution of 320×240 pixels (noncooled focus plane array, ATIR-M301, AG Digital Technology Corp, Taiwan, China). The properties of thermal cameras have been described in previous research.[22]

Possible IRTI device errors were controlled as follows. The IRTI detection room was indirectly ventilated and strictly temperature-controlled (ambient temperature: 23 °C ± 2 °C; relative humidity: 60%), as technically required to control errors arising from the surroundings. All subjects were instructed to remove clothing from the lower extremities and to rest for 10–15 minutes before IRTI testing, because clothing covers the body's infrared radiation and activity increases infrared radiation. The infrared camera was turned on approximately 30 min prior to the onset of testing. Physical examination was performed to identify conditions that might adversely affect IRTI: severe varicose veins, wounds, previous surgery, physical therapy, or electromyography. Locations with such conditions were avoided when possible to reduce errors derived from the participants. Possible errors derived from the device and from participants were thus minimized and kept stable.

IRTI was performed on the lower extremities to capture anterior and posterior standing views [Fig. 1(A)] and/or dorsal views of the extremities from the standing position and supine and genuflex supine positions [Fig. 1(B)]. Nonstanding DVT

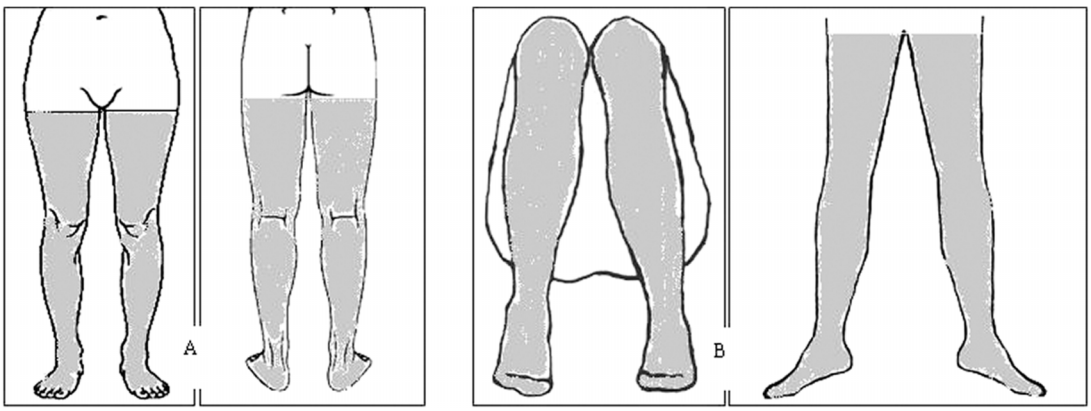


FIG. 1. Captured IRTI visual fields (shaded area). (A) Standing position. (B) Supine and genuflex supine position.

patients were required to lie on their backs; others received IRTI examinations in a standing position. In all subjects, the knee or popliteal fossa was set as the center of the visual capture field, which covered the entire lower extremity.

The DVT limb was the limb in which the DVT was located. All data collected were translated into grayscale digital images with IRTI analytical software for qualitative pseudocolor analysis (specific color-temperature responses) and quantitative temperature analysis. This system evaluates and analyzes changes in body surface temperature according to the IRTI presentation and distribution characteristics (PDCs), with different colors and temperature values. Because DVT can be located in the front or back of the limb, and because there are great temperature differences (TDs) between front and back, we predesigned regions of interest (ROIs). The ROI of the DVT was the area with the highest temperature in thermal imaging. Each DVT patient was definitively diagnosed; the ROI for each DVT patient was hand-drawn and compared with the results of CPUS and/or angiography. There were two types of ROI: (1) area-ROI, a specially shaped polygon, with sides just within the lateral and medial margins of the lower extremity, corresponding to the region of the DVT-affected veins, which displayed the highest temperatures in the captured IRTI, and (2) linear-ROI, a vertical line in the center of the area-ROI passing through the centers of the top and bottom edges of the area-ROI (Fig. 2). Because the surface temperatures of a normal adult body are usually comparable on the right and left sides, the IRTI PDCs were bilaterally symmetrical. In the present study, the ROI was selected where surface temperatures were highest and were not consistent with the opposite limb. The anterior knee is mainly composed of bony articulations without blood vessels. In contrast, the posterior knee area contains the popliteal vein, the great and small saphenous veins, and the vena cruralis and fibular veins, which drain into the popliteal vein. For this reason, the ROI was limited to below the knee in anterior views and included part of the supragenual area in posterior views.

The IRTI analysis software extracted the mean temperatures of area-ROI (T_area) and of linear-ROI (T_line) from captured images. The bilateral TDs in the T_area (TD_{area}) was defined as T_area on the DVT side minus T_area on the control side, and the TDs for T_line (TD_{line}) was defined as T_line on the DVT side minus T_line on the control side. The temperature values extracted from both linear-ROI and area-ROI were relative temperatures, but the TDs was an absolute temperature. Considering TDs is an absolute temperature and the infrared radiation captured by the IRTI device at every point on the subject is the same, TDs was designed to control temperature variations among and within patients in the research.

Because the temperature of area-ROI was the temperature of the entire DVT region, area-ROI included the temperature of all points in this area. However, the temperature of linear-ROI included only points on the vertical line. Thus, the temperature of linear-ROI was more accurate and closer to that of the DVT-affected vein than the temperature of area-ROI.

The IRTI PDCs of the DVT in all patients were classified into negative and positive subgroups. TDs less than 0.5 °C and absence of DVT-PDCs as mentioned above were defined as PDCs-negative; otherwise cases were defined as PDCs-positive.

2.C. Statistical analysis

Statistical analysis was performed using SPSS (PROC MIXED in SPSS 16.0 evaluation version, SPSS, Inc., Chicago, IL). As described above, T_area and T_line were extracted with IRTI analysis software. All data were expressed as mean ± standard deviation. Comparative statistical analyses of the differences in T_area and T_line between the DVT and non-

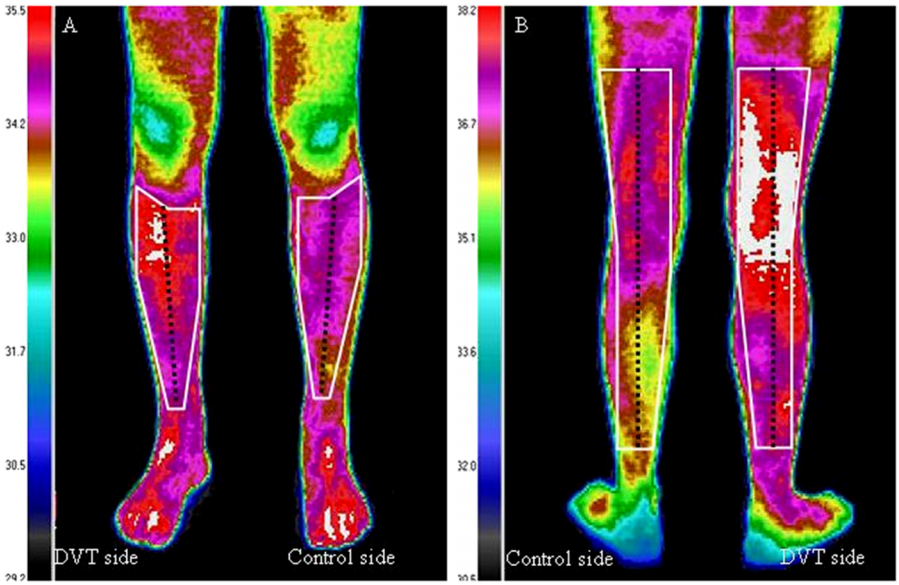


FIG. 2. IRTI detection range. The extracted ROI indicates the DVT location. The irregular white shapes indicate area-ROI and the black dotted lines indicate linear-ROI. (A) Anterior standing position. (B) Posterior standing position.

DVT limb in each DVT patient were performed with a paired t-test. Comparison of TD_{area} and TD_{line} between DVT patients and non-DVT volunteers was performed with a two-sided t-test. P values less than 0.05 and 0.01 were considered significant and highly significant, respectively.

3. RESULTS

3.A. Basic information about the participants

Sixty-four DVT patients (18–86 yr of age; mean age, 60.55 ± 18.07 yr; 30 women, 34 men) were included in this study. The underlying diseases or conditions that caused DVT in these 64 patients are shown in Table I. Sixty-four volunteers without DVT (20–84 yr of age; mean age, 60.42 ± 10.78 yr; 30 women, 34 men) were enrolled as the control group.

TABLE I. Patients with confirmed lower extremity DVT.

Underlying diseases	Number of patients
Malignancy	10 (15.63%)
Chronic obstructive pulmonary disease	10 (15.63%)
Postoperation	8 (12.5%)
Trauma, fracture, contusion, rupture	8 (12.5%)
Immobilization or obviously less activities	4 (6.25%)
Cardiovascular and cerebrovascular diseases	3 (4.69%)
Thrombophilia	3 (4.69%)
Significantly lower extremity varicose veins	2 (3.13%)
Long-haul air travel	1 (1.56%)
Rheumarthritis	1 (1.56%)
No predisposing factors	12 (18.75%)

3.B. IRTI PDCs of healthy volunteers and DVT patients

The IRTI PDCs in normal adult limbs was typically bilaterally symmetrical [Figs. 3(A) and 3(B)]. The TD_{area} range was 0.11 °C–0.28 °C (mean, 0.21 °C ± 0.05 °C) and the TD_{line} range was 0.15 °C–0.31 °C (mean, 0.22 °C ± 0.05 °C). There were no significant differences in lower extremity (thigh and calf) temperatures in either T_area or T_line in non-DVT volunteers.

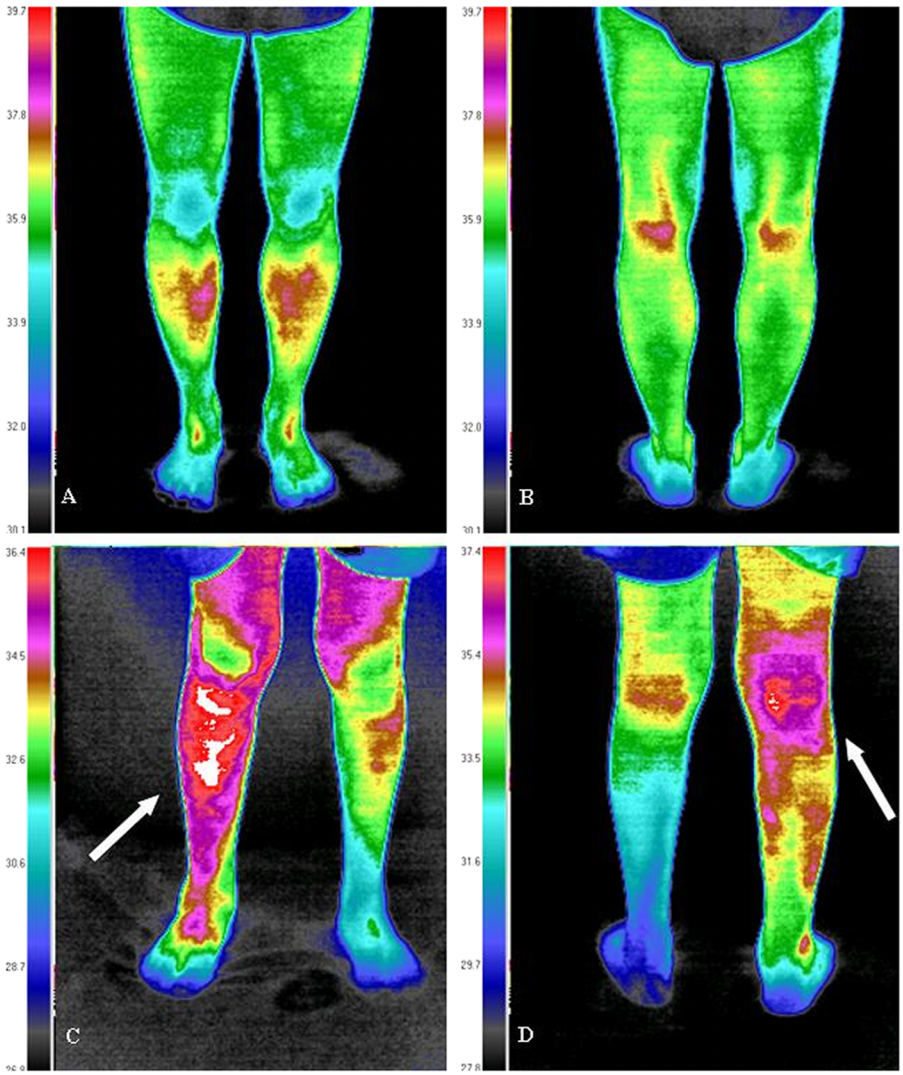


FIG. 3. IRTI PDCs in healthy volunteers and DVT patients. (A) and (B) Healthy volunteers. (C) and (D) DVT patients.

TABLE II. Bilateral temperatures and paired t-test results in DVT patients ($n = 64$, mean ± standard deviation, °C).

Variable 1	Variable 2	DVT side	Control side	p value
DVT patients	T_area	34.50 ± 1.77	33.64 ± 1.70	0.006
	T_line	34.81 ± 1.89	33.78 ± 1.74	0.002

In DVT patients, specific color-temperature responses showed that the IRTI PDCs on the DVT side was stronger (higher temperature) than on the control side. The IRTI PDCs in the ROI on the DVT-affected side was conspicuously abnormal, showing an asymmetrical heat zone. There were statistically significant differences between the DVT and non-DVT sides in both T_area ($P = 0.006$) and T_line ($P = 0.002$) in the DVT group ($P < 0.01$) (Table II).

Moreover, there were significant differences in the mean TDs between the DVT and non-DVT limbs for both T_area (TD_{area}) ($P = 2.55 \times 10^{-10}$) and T_line (TD_{line}) ($P = 1.27 \times 10^{-11}$) compared with those in non-DVT volunteers (Table III).

3.C. Agreement between IRTI and CPUS findings

According to IRTI testing, 62 (96.88%) of the 64 DVT patients were PDCs-positive and only two (3.13%) were PDCs-negative. Of these 62 PDCs-positive patients, 53 (82.81%) had satisfactory manifestations of IRTI PDCs in the DVT regions that agreed with positive results on CPUS. In nine patients (14.06%), IRTI PDCs did not definitively agree with the DVT regions as determined with CPUS, but still correctly indicated the DVT limb. The total diagnostic concordance rate was 96.88%. The results of IRTI, CPUS, and angiography are shown in Table IV.

4. DISCUSSION

DVT is associated with high blood viscosity and stagnated or blocked flow in the local blood vessels of the limb, resulting in tissue swelling and local accumulation of heat. These conditions can result in thrombosis, which further increases local temperature and causes the development of collateral circulation.[22] Limb temperature can be increased intrinsically in DVT patients because of vasodilatory substances, which are released as a result of the thrombotic process. Vessel walls can also secrete these substances before thrombus formation.[17] Altered blood flow in a DVT-affected limb can lead to changes in local infrared radiation at the DVT site. IRTI may be used to indirectly assess blood flow visually and in a dynamic manner and can be a simple, effective, and novel tool in adjunctive diagnostic screening for DVT.

TABLE III. Bilateral TDs (DVT side − control side) and two-sided t-test in non-DVT controls and DVT patients ($n = 64$ in each group, mean ± standard deviation, °C).

Variable	Non-DVT volunteer	DVT patients	p value
TD_{area}	0.19 ± 0.15	0.86 ± 0.71	2.55×10^{-10}
TD_{line}	0.21 ± 0.17	1.03 ± 0.79	1.27×10^{-11}

TABLE IV. Agreement between IRTI and CPUS findings ($n = 64$, %).

DVT regions detected by CPUS	IRTI PDCs-positive		IRTI PDCs-negative
	DVT regions	DVT limb	
Myenteric vein ($n = 10$)	5 (50%)	4 (40%)	1 (10%)
Tibial or fibular vein ($n = 18$)	14 (77.78%)	3 (16.67%)	1 (5.56%)
Popliteal vein ($n = 3$)	3 (100%)	—	—
Femoral vein ($n = 22$)	20 (90.91%)	2 (9.09%)	—
Iliac vein ($n = 11$)	11 (100%)	—	—
Total ($n = 64$)	53 (82.81%)	9 (14.06%)	2 (3.13%)

Even though the sensitivity and specificity of thermography are lower than those of CPUS, contact thermography with liquid crystal screens has been shown to be a suitable, inexpensive, simple, rapid, and noninvasive screening tool for DVT detection.[12–17] Thermography can detect clinically unrecognized thrombosis, identify patients with suspected unilateral lower-limb DVT, and diagnose DVT in high-risk patients.[9,12–17] Noncontact IRTI has been found effective as an adjunctive diagnostic screening test in DVT rabbit models.[22] In this study, the specific color-temperature responses showed that the bilateral IRTI PDCs in those without DVT was typically symmetrical. Quantitative analysis showed that the TDs value was a more accurate measurement than bilateral temperature for detecting DVT, because TDs was an absolute temperature whereas bilateral temperature was a relative temperature. Because the TDs between bilateral sides was below 0.3 °C, and was usually ±0.2 °C, we used 0.5 °C as a cutoff temperature in this study. Differences greater than this were not observed in the corresponding regions in non-DVT volunteers or on the unaffected side in the DVT patients. Differences greater than 0.5 °C may be attributed to slow blood flow in the DVT-affected regions, leading to significant findings in adjunctive diagnostic screening with IRTI.

IRTI and CPUS or angiography findings were consistent in DVT regions in 53 cases and in DVT limbs in 9 cases (Table IV). What accounted for disparities between IRTI PDCs and CPUS or angiography? Because the mechanisms of IRTI and CPUS are different, CPUS determines the exact location of the affected vessel, whereas IRTI reflects patterns of blood circulation. IRTI indicates the location where a DVT-affected vein is present. Greater consistency between CPUS or angiography findings and IRTI PDCs findings tended to be seen when the larger veins, such as the iliac vein, were affected. Severe varicose veins, surgery, skin wounds, and nonstanding positioning of the patient affected the DVT-PDCs findings as well (Table V). As a result, in most cases of lower limb DVT, IRTI yielded consistent findings at the same DVT regions as CPUS or angiography. Moreover, we observed that the TD_{area} and TD_{line} in DVT patients were significantly different from those in non-DVT volunteers (Table III). The TD_{area} and TD_{line} on the DVT-affected side were significantly higher than those

TABLE V. Reasons for discordant findings in nine patients.

Location	Number	Reasons
Myenteric vein	2	PDCs were expressed on foot where the located venous was smaller than the distal venous.
	2	PDCs were expressed on popliteal where the PDCs of located venous were enlarger.
Tibial or fibular vein	2	Nonstanding patients, with position could affect the IRTI result.
	1	After joint replacement surgery, the expressed PDCs was fold with the surgical wound area, where could be identified as highly doubtful DVT.
Femoral vein	2	Patients had very severe varicose veins. IRTI can clearly distinguish mild and moderate varicose veins, but very severe varicose veins could be identified as highly doubtful DVT.

on the normal side and than those in non-DVT volunteers (Table III).

Our findings indicate that abnormal body surface temperature in a lower limb suggests abnormal blood flow in that area. Slow blood flow probably indicates obstructed blood supply or thrombosis in the embolic region. Given the similar temperature of the skin and the limb vessels, assessment of skin temperature is useful for indirect diagnosis of vascular diseases in the limbs. Peripheral arterial diseases are typically associated with ischemia, while peripheral venous diseases are generally congestive. In view of the close relationship between infrared radiation and blood flow volume, it is not surprising that abnormal hypothermia is commonly detected by IRTI in peripheral arterial diseases,[15,24] while local hyperthermia is seen with IRTI in DVT patients, as found in our study.

Ambient temperature is important when conducting IRTI testing, because it affects body temperature measurement, interfering with the background of the thermal imaging. Therefore, the ambient temperature in this study was standardized at 23°C ± 2°C. Participants are also more comfortable in standardized room temperature. To reduce possible errors, a single room was used for testing, with steady ambient temperature and humidity. This standardization minimized the impact of these variables on body temperature measurements.

However, this study has several limitations. Many factors can interfere with the discrimination and analysis of IRTI, including skin wounds, patient positioning, activity, ambient temperature, and clothing and those factors leaded to certain heterogeneities in the patients included. This study also had several deficiencies in study design that limits its clinical importance such as noncontrolled, nondouble-blinded, and with no comparison to CPUS or angiography. Nevertheless, the present study was a preliminary and early attempt to explore the possible diagnostic role of IRTI in lower limb DVT patients. Future studies should address these limitations and preferably include a larger sample size of subjects which enable stratifying the subjects by normal controls, patients with or without skin wounds in different positionings. Hopefully, this will adequately validate our findings.

In summary, we found that IRTI PDCs and IRTI TDs values could effectively detect lower limb DVT. As a simple, noncontact, economical, and rapid test, IRTI can be used as an adjunct to CPUS in diagnostic screening of clinically suspected DVT cases.

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[1]V. Spiss *et al.*, "Contrast enhanced ultrasound of the lower limb deep venous system: A technical feasibility study. Technical innovation," Med. Ultrason. **13**(4), 267–271 (2011).
[2]I. Datta *et al.*, "Complications related to deep venous thrombosis prophylaxis in trauma: A systematic review of the literature," J. Trauma Manage Outcomes **4**, 1–11 (2010).
[3]S. Brian, B. Fries Richard, and S. Bhagwan, "Clinical characteristics of patients with isolated calf vein thrombosis in a large teaching hospital," Int. J. Vasc. Med. **2011**, 414093 (2011).
[4]T. Zhu, I. Martinez, and J. Emmerich, "Venous thromboembolism: Risk factors for recurrence," Arterioscler., Thromb., Vasc. Biol. **29**(3), 298–310 (2009).
[5]R. Kreidy *et al.*, "Value of venous color flow duplex scan as initial screening test for geriatric inpatients with clinically suspected pulmonary embolism," Vasc. Health Risk Manage. **7**, 585–589 (2011).
[6]Z. F. Wang *et al.*, "Acute pulmonary embolism after pulmonary lobectomy," J. Thorac. Dis. **4**(1), 76–82 (2012).
[7]A. de Oliveira *et al.*, "Duplex scan in patients with clinical suspicion of deep venous thrombosis," Cardiovasc. Ultrasound **6**, 53–57 (2008).
[8]M. T. Rondina *et al.*, "(18)F-FDG PET in the evaluation of acuity of deep vein thrombosis," Clin. Nucl. Med. **37**(12), 1139–1145 (2012).
[9]H. S. Sajjadi, "A systematic review on the effectiveness of thermography in diagnosis of Diseases," Int. J. Imaging Syst. Technol. **23**(2), 188–193 (2013).
[10]A. L. Herrick and C. Hutchinson, "Vascular imaging," Best Pract. Res Clin. Rheumatol. **18**(6), 957–979 (2004).
[11]J. Kang *et al.*, "Study on improving blood flow with Korean red ginseng substances using digital infrared thermal imaging and doppler sonography: Randomized, double blind, placebo-controlled clinical trial with parallel design," J. Tradit. Chin. Med. **33**(1), 39–45 (2013).
[12]R. Pochaczevsky, G. Pillari, and F. Feldman, "Liquid crystal contact thermography of deep venous thrombosis," Am. J. Roentgenol. **138**(4), 717–723 (1982).
[13]A. Kohler *et al.*, "Diagnostic value of duplex ultrasound and liquid crystal contact thermography in preclinical detection of deep vein thrombosis after

proximal femur fractures," Arch Orthop. Trauma Surg. **117**(1-2), 39–42 (1998).
[14]H. Bounameaux *et al.*, "Value of liquid crystal contact thermography and plasma level of D-dimer for screening of deep venous thrombosis following general abdominal surgery," Thromb. Haemost. **67**(6), 603–606 (1992).
[15]E. W. Cameron *et al.*, "Liquid crystal thermography as a screening test for deep vein thrombosis in patients with cerebral infarction," Eur. J. Clin. Invest. **21**(5), 548–550 (1991).
[16]R. Hoffmann, F. Largiadèr, and H. P. Brütsch, "Liquid crystal contact thermography–A new screening procedure in the diagnosis of deep venous thrombosis," Helv. Chir. Acta. **56**(1-2), 45–48 (1989).
[17]D. A. Sandler and J. F. Martin, "Liquid crystal thermography as a screening test for deep-vein thrombosis," Lancet **325**(8430), 665–668 (1985).
[18]L. M. Katz *et al.*, "Infrared imaging of trauma patients for detection of acute compartment syndrome of the leg," Crit Care Med. **36**(6), 1756–1761 (2008).
[19]C. L. Herry and M. Frize, "Quantitative assessment of pain-related thermal dysfunction through clinical digital infrared thermal imaging," Biomed. Eng. Online **3**(1), 19–32 (2004).
[20]S. Bagavathiappan *et al.*, "Infrared thermal imaging for detection of peripheral vascular disorders," J. Med. Phys. **34**(1), 43–47 (2009).
[21]S. Bagavathiappan *et al.*, "Investigation of peripheral vascular disorders using thermal imaging," Br. J. Diabetes Vasc. Dis. **8**(2), 102–104 (2008).
[22]F. Deng *et al.*, "Infrared thermal imaging as a novel evaluation method for deep vein thrombosis in lower limbs," Med. Phys. **39**(12), 7224–7231 (2012).
[23]N. Arora *et al.*, "Effectiveness of a noninvasive digital infrared thermal imaging system in the detection of breast cancer," Am. J. Surg. **196**, 523–526 (2008).
[24]J.-Y. Park, "The effectiveness of digital infrared thermographic imaging in patients with shoulder impingement syndrome," J. Shoulder Elbow Surg. **16**(5), 548–554 (2006).